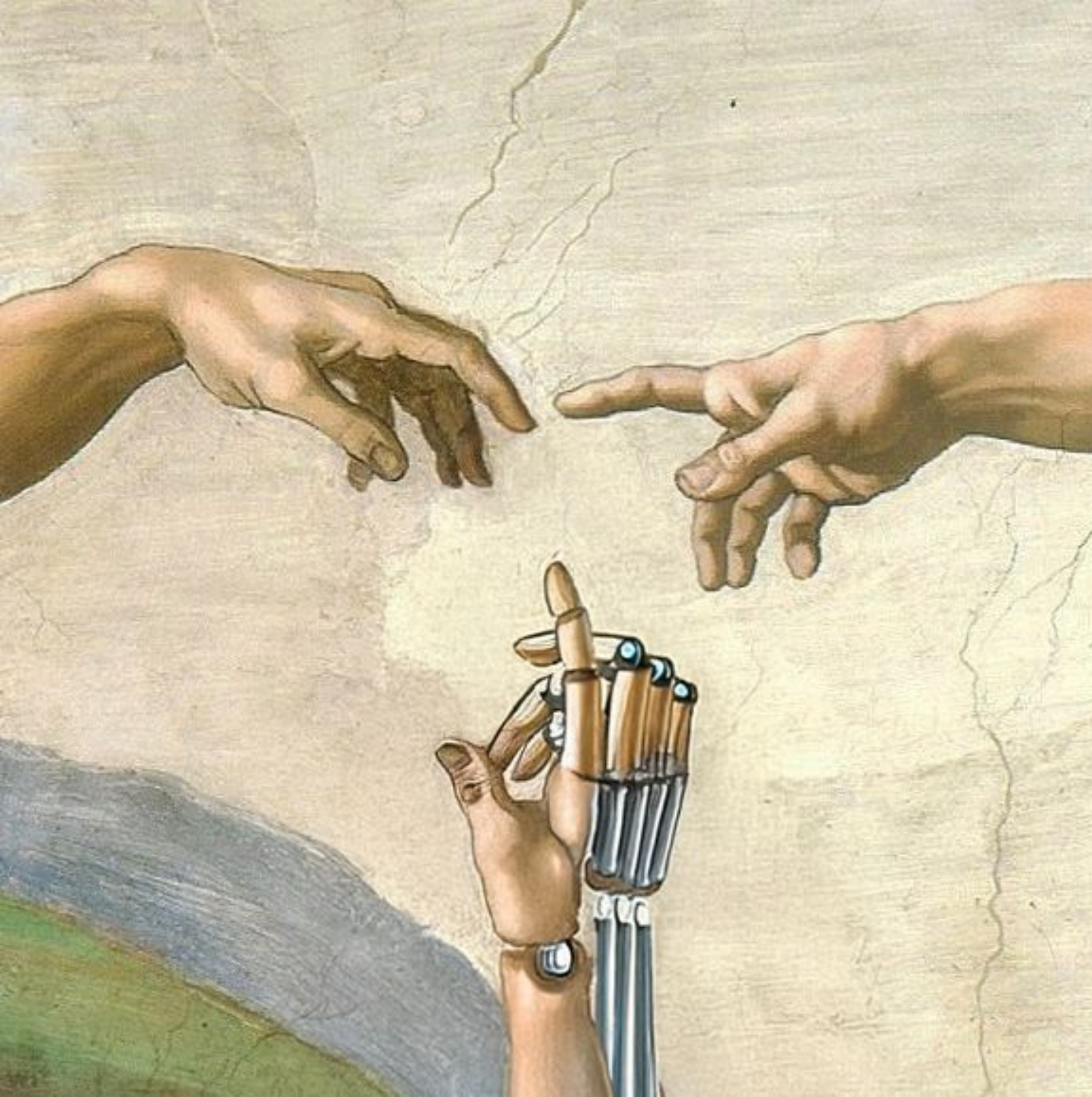




Neural Networks and Deep Learning

Dr. Umer

Introduction

EDUCATION

Doctorate (PhD) in Computer Science from TUM, Germany

School of Computation, Information and Technology at the Technical University of Munich (TUM)

Dissertation: Leveraging Representation Learning for Multi-Modal and Graph-Structured Data

Masters in Communication Engineering from TUM, Germany

Thesis: Learning to Rank E-Commerce Products with Multiple Noisy Feedback Signals Extracted from User Interactions

Bachelors in Electrical Engineering from NUST, Pakistan

EXPERIENCE



AI Product Owner at Unite

March, 2025 to Present

Core Focus: Using AI to understand & enhance content and to enable Unite to make better decisions



AI Project Lead, Government Collaborative Research, Germany

Jan, 2024 to Present

Core Focus: Using AI to Develop the Product Content Graph and ensuring efficient customer support



Senior Data Scientist at Unite, Munich, Germany

April, 2022 to Dec, 2023

Analyzed large volumes of data, built prototypes and collaborated with stakeholders to derive insights and deliver outcomes



Applied Research Scientist at Mercateo Group, Germany

Dec, 2016 to Mar, 2022

Worked with the Analytics Team as Machine Learning Researcher

Welcome to Analytics CUE (simply, Q)- Content Understanding and Enhancement



Q (James Bond) - generated by ChatGPT



Student Name	Topic	Type
Xiaoyu Zhang	Leveraging LLMs to judge the cooccurrence of labels for products	Internship
Maram Bayadsi	Few-Shot Text Classification for E-Commerce Data	Masters Thesis
Maro Grigoryan	Variational Methods for Graph Representation Learning	Masters Thesis
Mohmmad Kashif Akhtar	Machine-Learning With Graph Neural Networks	Masters Thesis
Leopold Wieser	Multi-Label Multi-Modal Classification with Auto-Encoders	Bachelor's Thesis
Zihui Pan	Using Graph Auto-encoders for Compositional Learning of Multimodal Data	Masters Thesis
Murtaza Raza	Compositional Learning of Text and Images	Internship
Mark Sandal	Heterogeneous Graph Representation Learning	Internship
Omran Kaddah	Community Detection using Graphs	Masters Thesis
Omran Kaddah	Unsupervised Disentangled Representation Learning: Survey on Methods of Learning Disentangled Representation with VAEs	IDP
Shyam Arumugaswamy	Metapath- and Entity-aware Graph Neural Network for Recommendation	Masters Thesis
Egor Labintcev	Retrieving Images based on Multimodal (Image-Text) Query	Internship + Masters Thesis
Abhilasha Mohania	Multi-Modal Classification with Attention	Internship
Simon Gasse	Convolutional Fusion for Multi-Modal Classification	Masters Thesis
Jonas Natzer	Improving Text Based Multi-Label Classification	Masters Thesis
Dmytro Rybalko	Mend The Learning Approach, Not the Data: Insights for Ranking E-Commerce Products	Masters Thesis

TEACHING AND MENTORING

Machine Learning Course at TUM “Information Retrieval in High Dimensional Data”

For 5 years: 2018 to 2023 at Technical University of Munich (TUM)

Tutored an ML course at TUM Information Retrieval in High Dimensional with 120+ students each year.

Lectures (Recorded due to Covid-19): <https://youtu.be/xdeelKqbA4s?si=NGp4OvWkYLmtgoNQ>

Recruiting and Mentoring of Masters Students

Mar, 2018 to Present

Recruited and Mentored over 20+ Master Students in the area of Variational Graph Methods, Graph Neural Networks, Computer Vision, NLP, Clustering and Multi-Modal Data Representation Learning

Course Overview

Pre-Requisites – Movies you have watched earlier

- Linear Algebra [Must Have Watched]
- Probability and Statistics [Must Have Watched]
- Information Theory [Nice to Have Watched]
- Numerical Computation [Nice to Have Watched]

PS: We will need to establish a 3-member censor board for this course.

Course Outline - Before the Interval

- Neural Networks Basics [Ch# 5]
- Deep Feedforward Networks [Ch# 6]
- Regularization for Deep Learning [Ch# 7]
- Optimization for Deep Learning [Ch# 8]
- Convolutional Neural Networks (CNNs) [Ch# 9]
- Things to consider in Practical Scenarios [Ch# 11/12]
- Autoencoders [Ch# 14]
- Generative Models, Variational Autoencoders [Ch# 14]

Course Outline - After the Interval

- Recurrent Neural Networks (RNNs) [Ch# 10]
- Attention and Transformer Architecture [Special]
- Ethics in AI, Bias and Fairness in Deep Learning [Special]
- Applications of Deep Learning [Special]
 - Approximate Inference [Ch# 19]
- Group Projects
- Supervision and Feedback Sessions
- Project Presentations
- Review of Key Learning Points, Future Directions

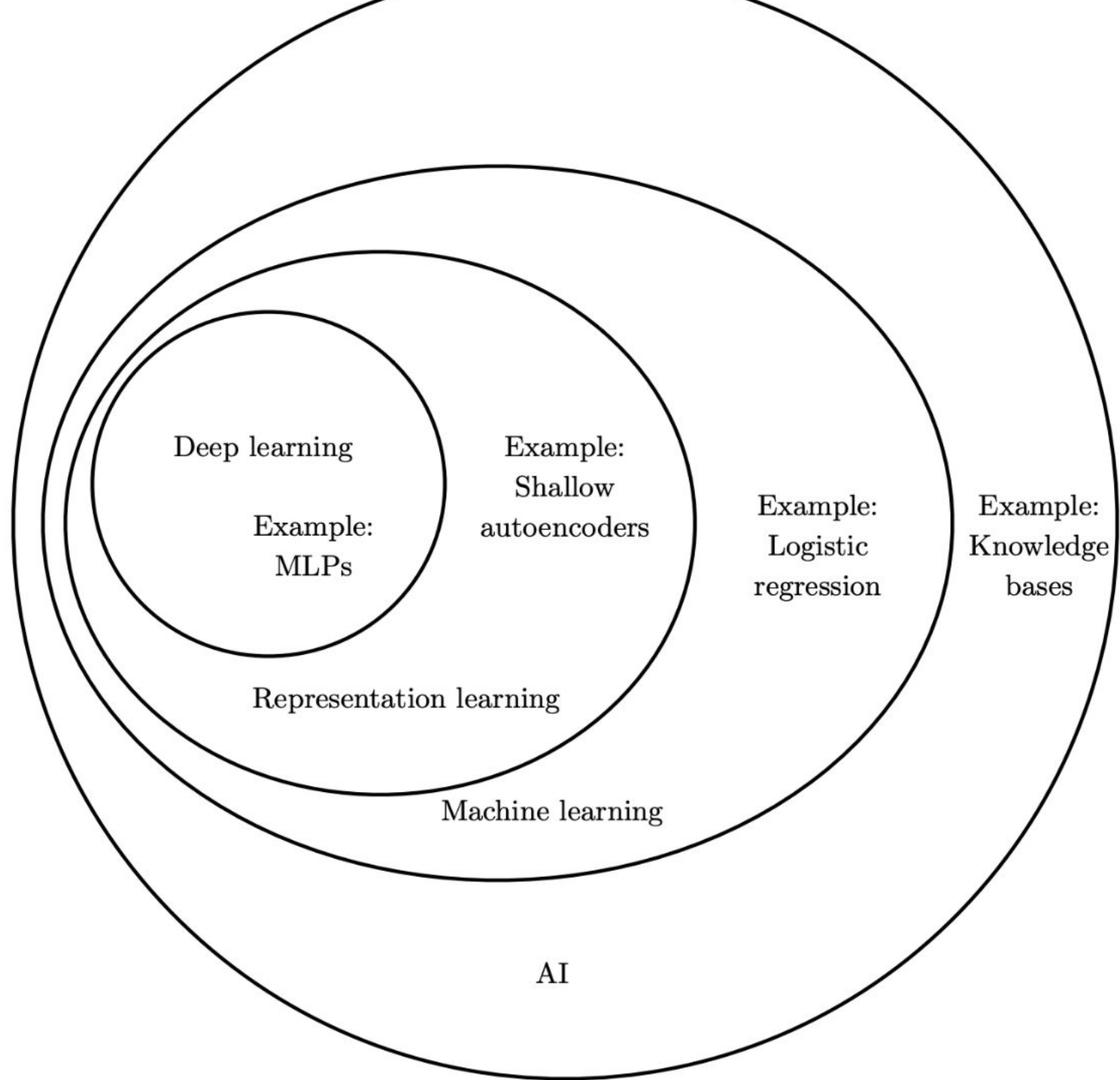
Grading Breakdown

• Quiz	20%	[Unannounced - Low Attendance]
• Mid Term	20%	[Announced - Medium Difficulty]
• Project	30%	[Announced - You Choose Difficulty]
• Final Term	30%	[Announced - Medium-Hard Difficulty]

Reference Book

- Goodfellow et al. <https://www.deeplearningbook.org/>

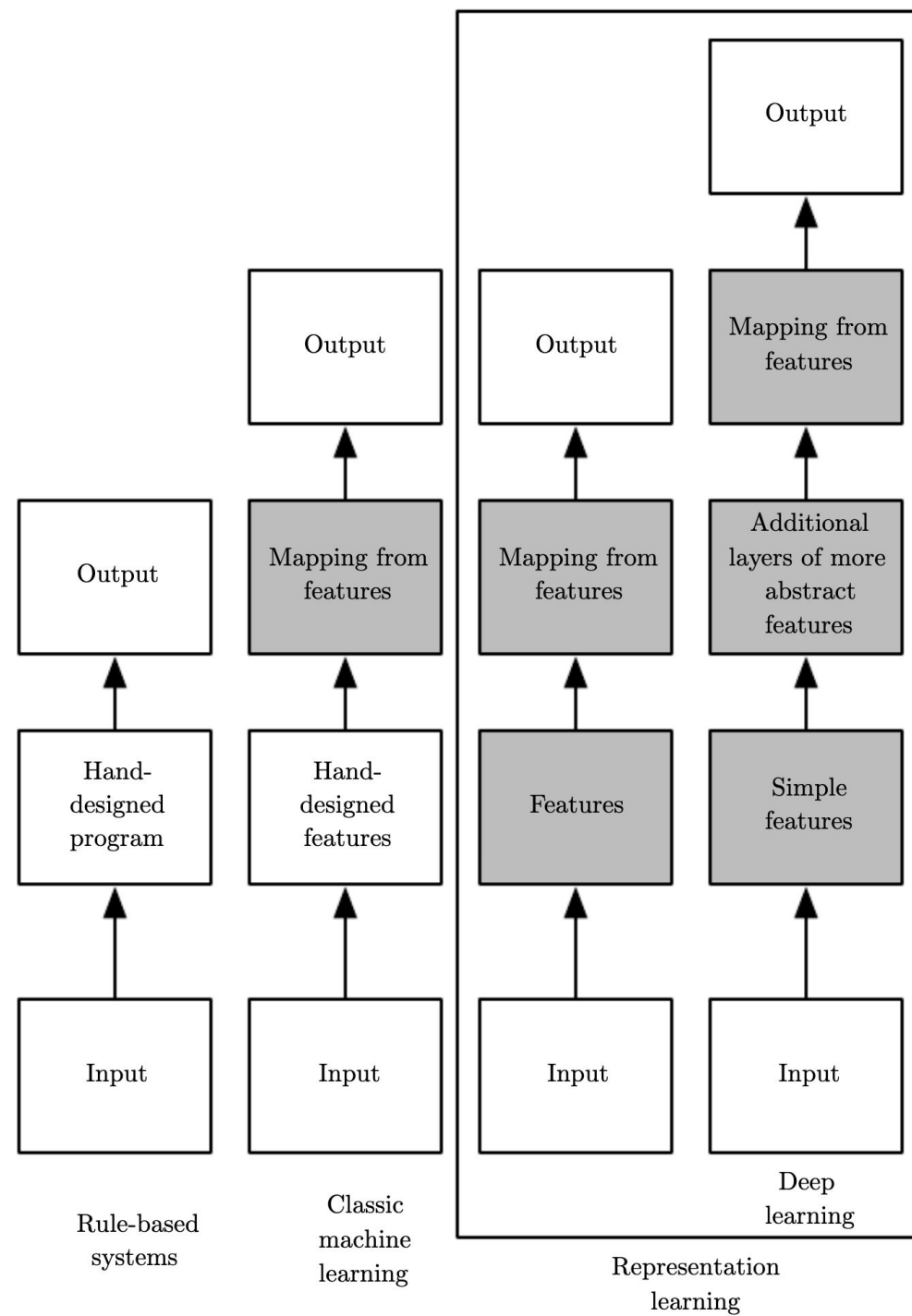
Let's Start



@source

{Goodfellow et al 2016

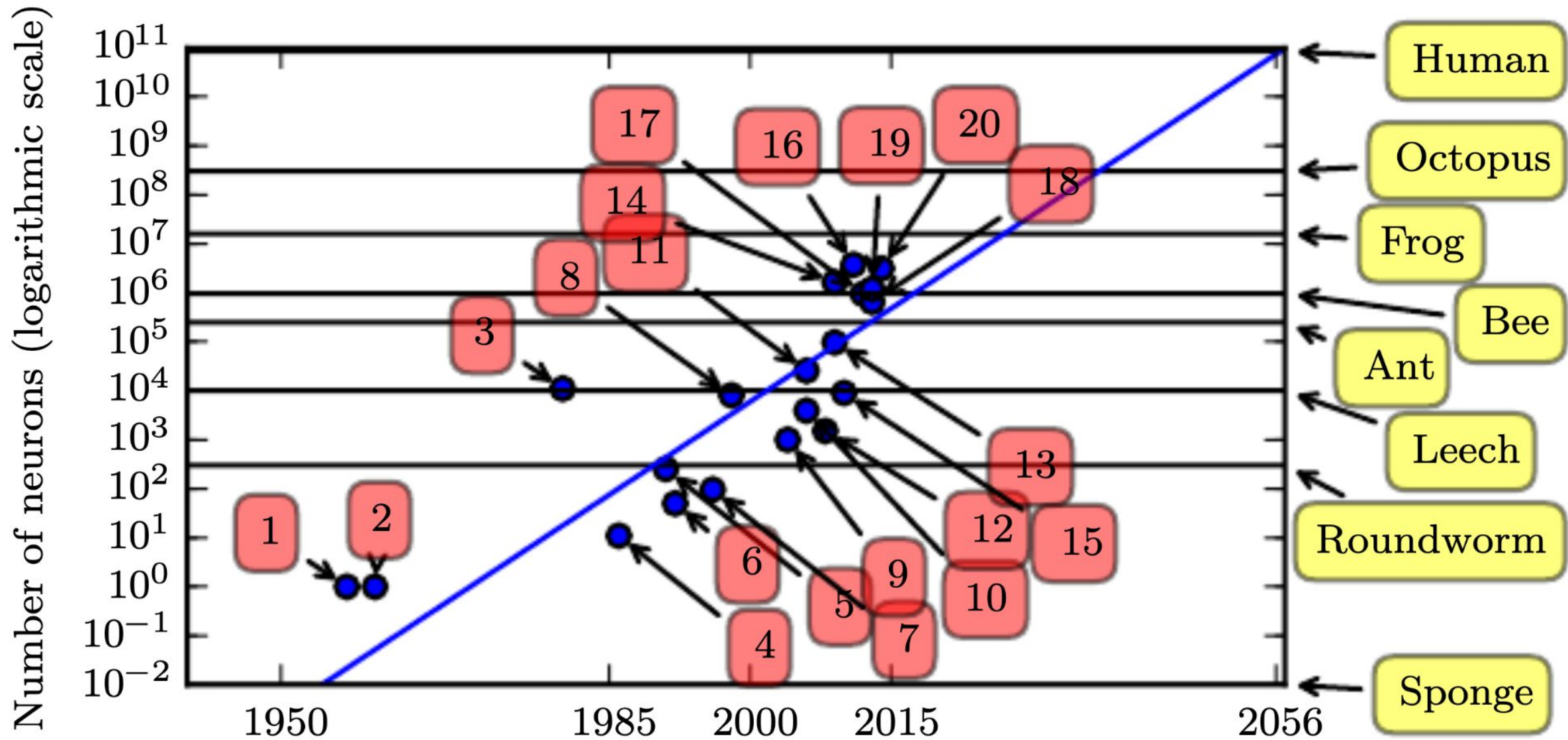
Book: [Deep Learning](#)}



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{Goodfellow et al 2016

Book: [Deep Learning](#)}



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{Goodfellow et al 2016

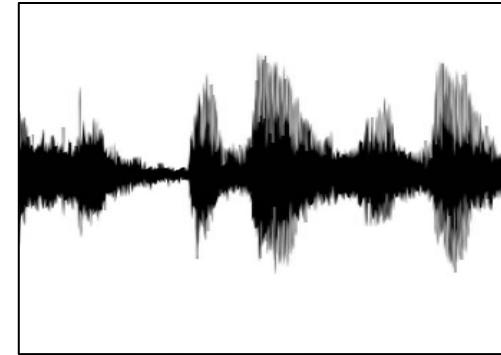
Book: [Deep Learning](#)}

Structured Data

Size	#bedrooms	...	Price (1000\$s)
2104	3		400
1600	3		330
2400	3		369
⋮	⋮		⋮
3000	4		540

User Age	Ad Id	...	Click
41	93242		1
80	93287		0
18	87312		1
⋮	⋮		⋮
27	71244		1

Unstructured Data



Audio



Image

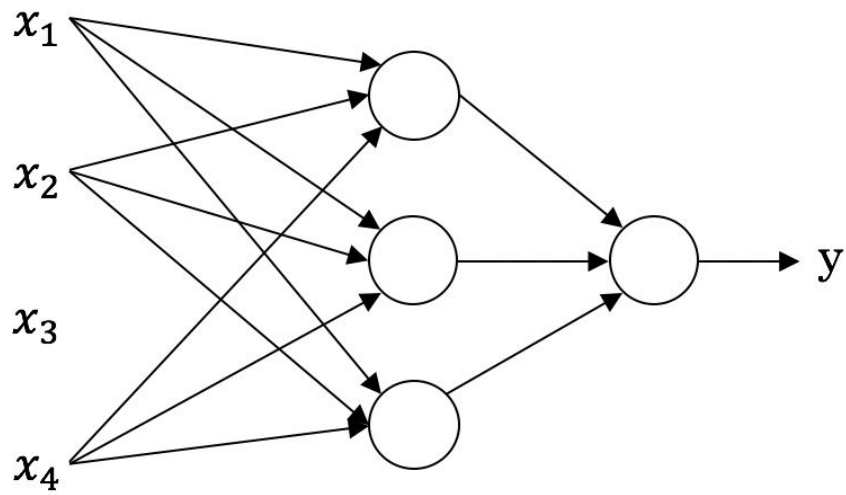
In a far far land, seven
years ago...

Text

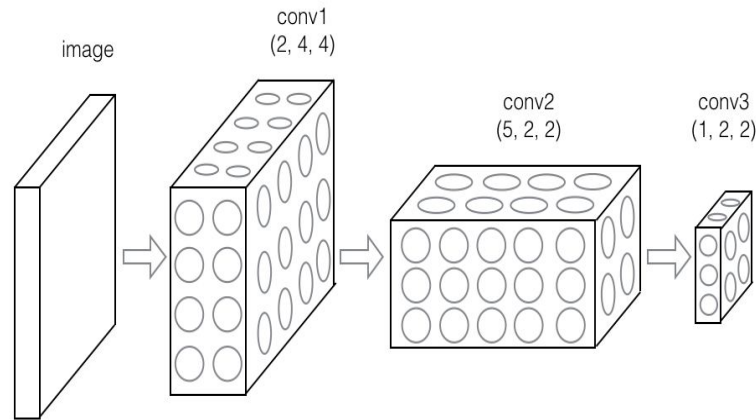
What is a Neural Network?

Best choice to learn functions due to
UAT [Universal Approximation Theorem]

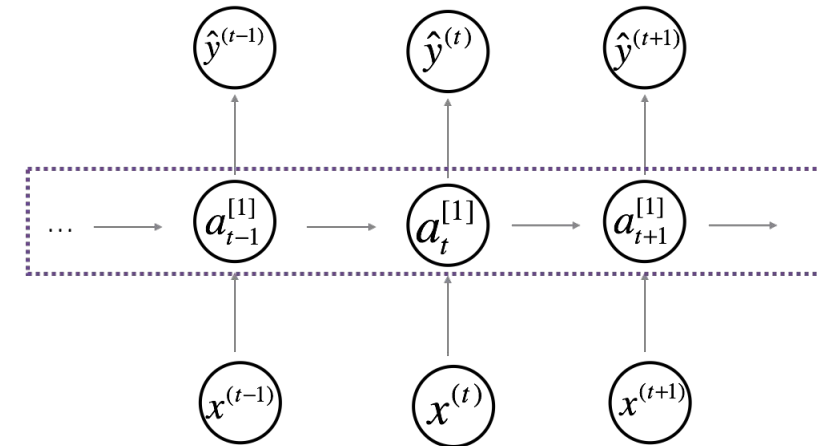
Neural Network examples



Standard FC NN

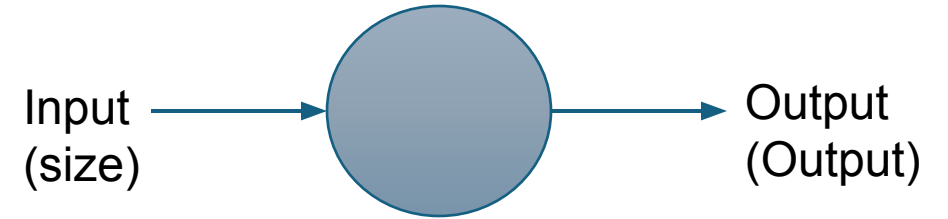
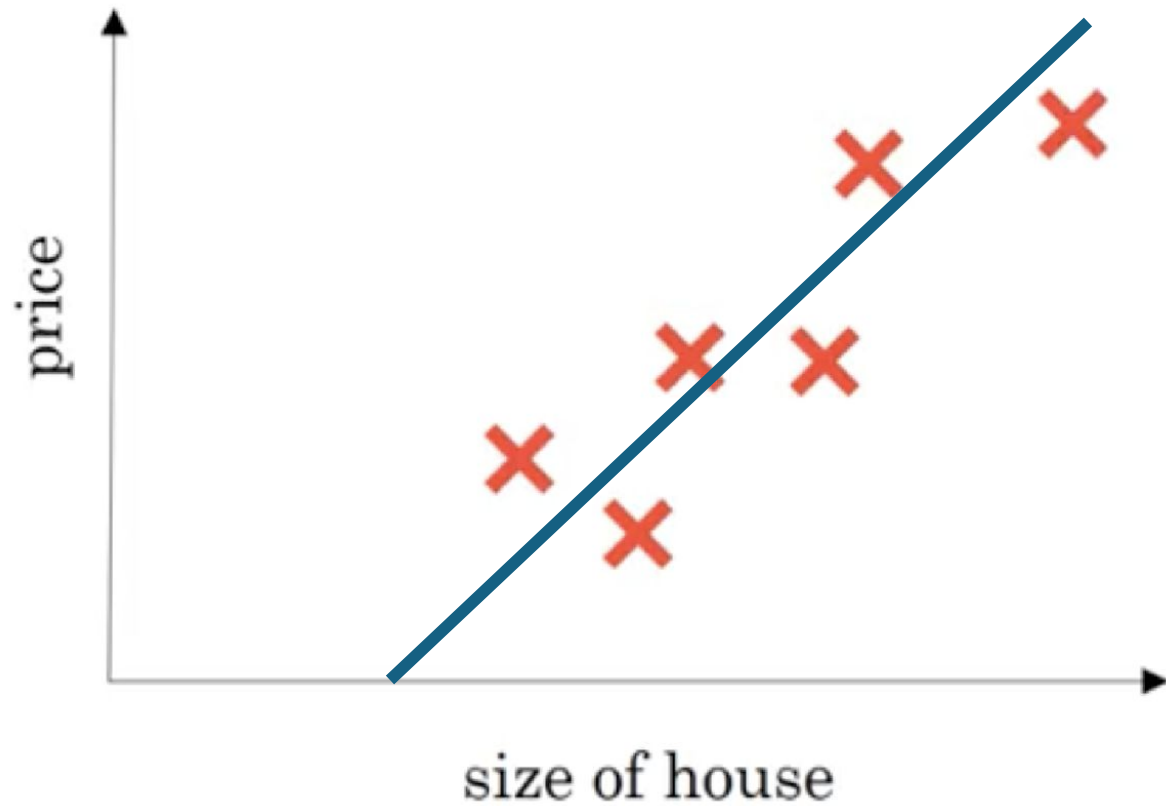


Convolutional NN

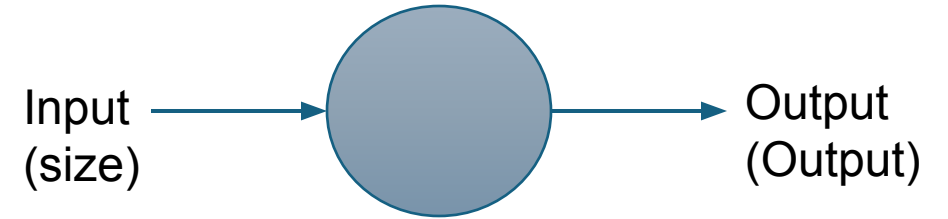
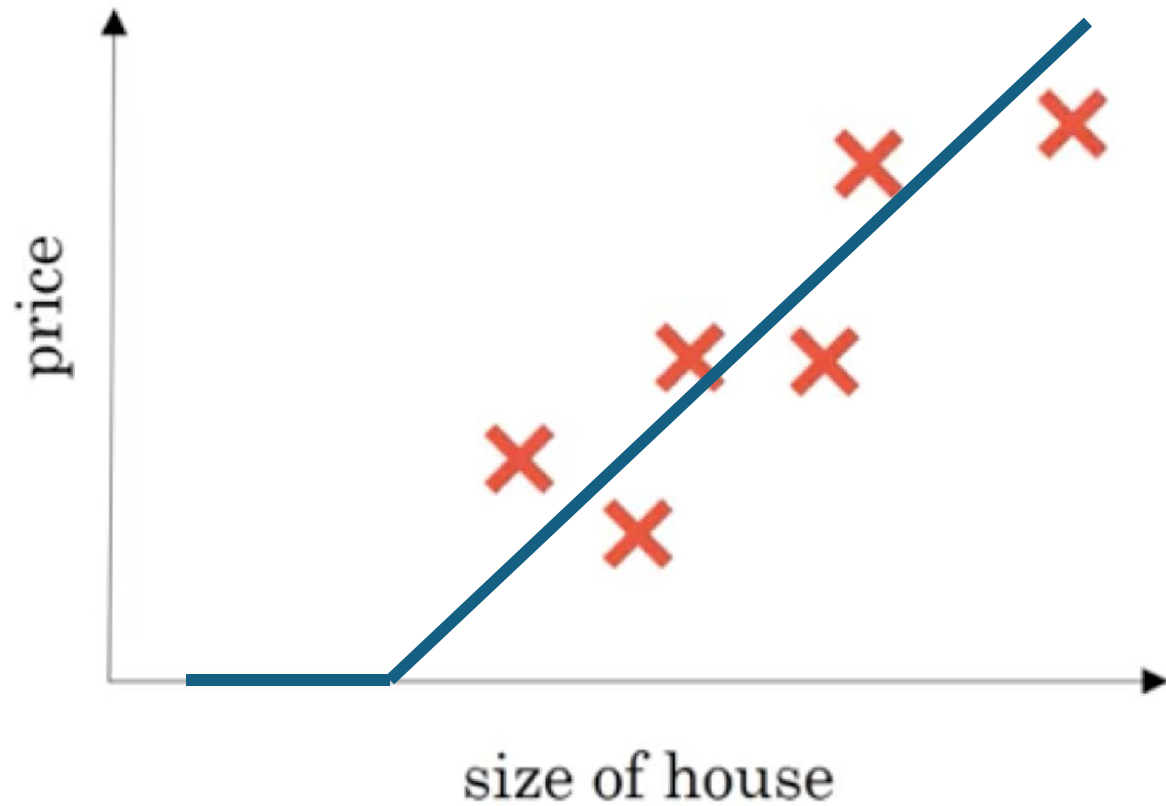


Recurrent NN

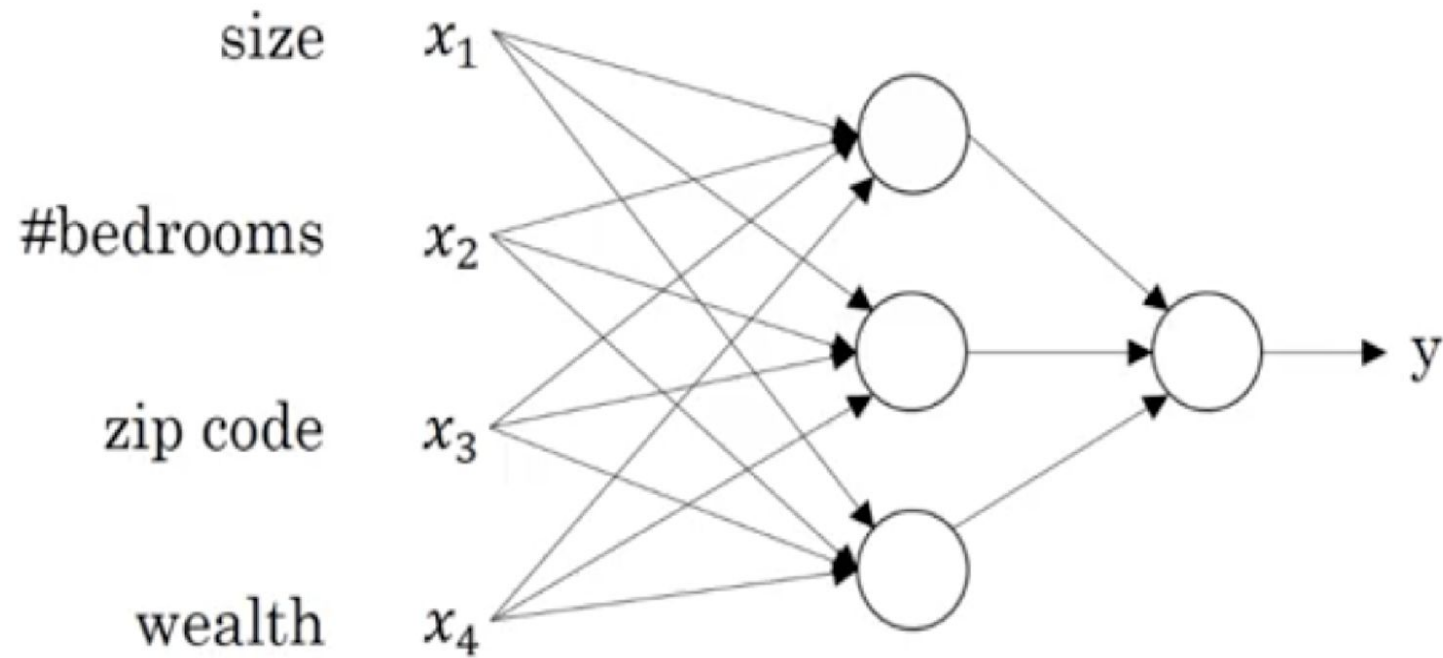
Housing Price Prediction



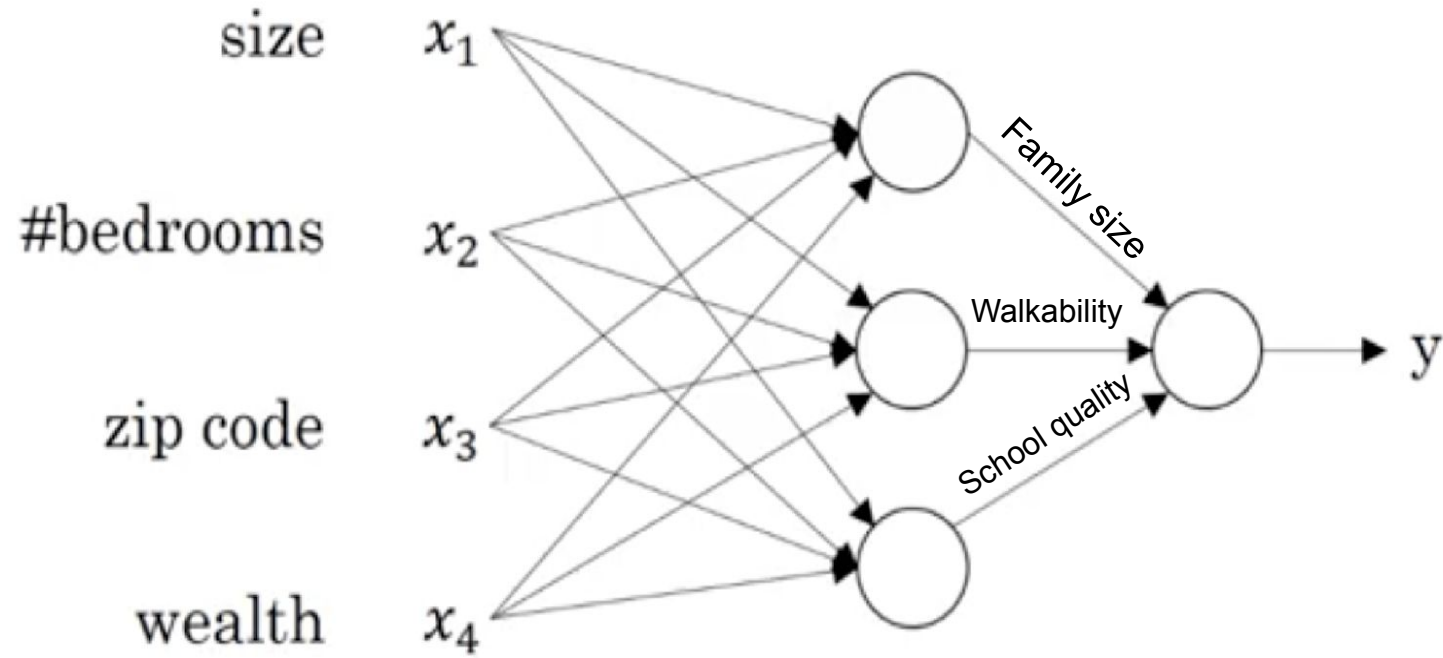
Housing Price Prediction



Housing Price Prediction



Housing Price Prediction



Equation of Single Neuron

- Standard Neural Network

$$y = f(wx + b)$$

Where

y = output

f = activation function

w = weight matrix

x = input vector

b = bias

Supervised Learning

Input(x)	Output (y)	Application	
Home features	Price	Real Estate	
Ad, user info	Click on ad? (0/1)	Online Advertising	
Image	Object (1,...,1000)	Photo tagging	
Audio	Text transcript	Speech recognition	
English	Chinese	Machine translation	
Image, Radar info	Position of other cars	Autonomous driving	

Thanks!